

guarantees that we have no duality gap. Then, we can see that γ is concentrated on the graph of $x \mapsto x - \nabla \phi(x) := \nabla u(x)$ provided u (or ϕ , which is equivalent) is differentiable μ -a.e. Since $\phi \in L^1(\mu)$ we infer that u is finite μ -a.e. and, since u is convex, we get $\text{spt}(\mu) \subset \{u < +\infty\}$, which is a convex set. Note that $\partial(\{u < +\infty\})$ can be expressed locally as the graph of a concave function, hence it is $(d-1) - C^2$ rectifiable set, and is μ -negligible by assumption. In the interior of $\{u < +\infty\}$, u is differentiable μ -a.e. (again, because of the assumption on μ and of u being convex). $\square$ $\square$

Remark 1.23. As a consequence of all these considerations, the quadratic case gives a very interesting result in dimension one. Suppose that $\mu \in \mathcal{P}(\mathbb{R})$ is atomless. Then every convex function is differentiable μ -a.e., since we know that the set of non-differentiability points of a convex function is at most countable (this is a consequence of the fact that, if ψ is convex, then the intervals $]\psi'_l(x), \psi'_r(x)[$, where ψ'_l and ψ'_r denote the left and right derivatives, are all non-empty and disjoint when x ranges among non-differentiability points). This implies the existence of an optimal transport map for the quadratic cost between μ and any measure $\nu \in \mathcal{P}(\mathbb{R})$. This transport map will be the derivative of a convex function, i.e. a non-decreasing map.

Remark 1.24. Some of the properties of the quadratic case stay true if we pass to a general cost $c(x, y)$ which is $C^{1,1}$ and satisfies the twist condition of Definition 1.16. Obviously, the fact that the transport is the gradient of a convex function is no more true; on the other hand, the differentiability properties of Kantorovich potentials are the same as for convex functions, since one can prove that ϕ is semi-concave (i.e. concave, up to subtracting a quadratic function), and this allows to use the same assumptions on μ as in Theorem 1.22.

1.3.2 The quadratic case on the flat torus

It is useful and interesting to see what happens if we replace measures on $\mathbb{R}^d$ with measures on the flat torus $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ and the cost $\frac{1}{2}|x - y|^2$ with the squared distance on the torus, i.e. $c(x, y) = \frac{1}{2}||x - y||^2$, where

$$||z|| := \min\{|z + k| : k \in \mathbb{Z}^d\}. \quad (1.5)$$

This is the simplest example of optimal transport on a manifold, for which a general existence theorem has been established by McCann in [231]. Yet, the case of the torus, first studied in [128], is much simpler, because of the structure that $\mathbb{T}^d$ inherits from $\mathbb{R}^d$. In many cases (we will see in Chapters 5 and 6, in particular), studying what happens on the torus allows to understand the main qualitative features of the problem, getting rid of difficulties arising from lack of compactness and/or boundary issues. In the case of compactly supported measures, it is possible to view their supports as part of a large cube, to be identified with a torus, and this at no price. Indeed, if the measures are far from the boundary of the cube, the optimal transport maps on the torus and on the Euclidean space will coincide.

Note that an optimal $\bar{k}$ in (1.5) always exists. If we take $z \in Q := [-\frac{1}{2}, \frac{1}{2}]^d$, then $\bar{k} \in \{-1, 0, +1\}^d$, and it is unique (and equal to 0) unless $z \in \partial Q$. Moreover, the function $\mathbb{R}^d \ni z \mapsto \frac{1}{2}||z||^2$ is semi-concave, as it is the infimum of a family of functions

with Hessian equal to the identity matrix I . It is differentiable at every point where the optimal $\bar{k}$ is unique. For a given point x , we call *cutlocus* of x the set of points y such that the squared distance function, viewed as a function of the pair (x, y) , is not differentiable at (x, y) . This is a general notion in differential geometry, and we denote it by $\text{cut}(x)$ (see [294] for its use in optimal transport), but in the case of the flat torus it simply coincides with the set of points y such that $y - x \in \partial Q + \mathbb{Z}^d$.

Theorem 1.25. *Take $\mu, \nu \in \mathcal{P}(\mathbb{T}^d)$, with $\mu \ll \mathcal{L}^d$ and $c(x, y) = \frac{1}{2}||x - y||^2$. Then there exists a unique optimal transport plan $\gamma \in \Pi(\mu, \nu)$, it has the form $\gamma = \gamma_\Gamma$ and the optimal map T is given by $T(x) = x - \nabla \varphi(x)$ for a.e. x , where the sum $x - \nabla \varphi(x)$ is to be intended modulo $\mathbb{Z}^d$. Here the function φ is a Kantorovich potential, solution to the dual problem, in the transport from μ to ν for the cost c . Moreover, for a.e. $x \in \mathbb{T}^d$, the point $T(x)$ does not belong to $\text{cut}(x)$.*

Proof. The strategy is the same as for the costs of the form $h(x - y)$. We take an optimal transport plan γ , together with a pair (φ, ψ) , solution of the dual problem, and we fix $(x_0, y_0) \in \text{spt}(\gamma)$. We have

$$\frac{1}{2}||x - y_0||^2 - \varphi(x) \geq \psi(y_0), \quad \frac{1}{2}||x_0 - y_0||^2 - \varphi(x_0) = \psi(y_0).$$

Now, take $k_0 \in \mathbb{Z}^d$ such that $||x_0 - y_0|| = |x_0 - y_0 - k_0|$ and observe that, from the inequality $||x - y_0|| \leq |x - y_0 - k_0|$, we have

$$\frac{1}{2}|x - y_0 - k_0|^2 - \varphi(x) \geq \psi(y_0), \quad \frac{1}{2}|x_0 - y_0 - k_0|^2 - \varphi(x_0) = \psi(y_0).$$

This means that the function $x \mapsto \frac{1}{2}|x - y_0 - k_0|^2 - \varphi(x)$ is minimal at $x = x_0$. Yet, $\varphi(x) = \inf_y \frac{1}{2}||x - y||^2 - \psi(y)$ is a semi-concave function, as it is defined as an infimum of uniformly semi-concave functions. Hence, it is differentiable a.e. and μ -a.e.

If we assume that x_0 is such that $\nabla \varphi(x_0)$ exists, we deduce from the minimality above that $x_0 - y_0 - k_0 = \nabla \varphi(x_0)$. This implies that k_0 is uniquely determined by x_0 and y_0 , and hence $y_0 \notin \text{cut}(x_0)$. Moreover, y_0 is uniquely determined by x_0 and we get the usual result for the existence and uniqueness of the optimal map. $\square \quad \square$

Remark 1.26. In the proof of the previous theorem we can also observe that we have $k_0 = 0$ if $x_0 - y_0 \in Q$; this means in particular that whenever $\text{spt}(\mu), \text{spt}(\nu) \subset \frac{1}{2}Q$, then the transport is completely identical to what we would have with the classical quadratic cost in $\mathbb{R}^d$ instead of using the periodic framework of $\mathbb{T}^d$.

1.4 Counter-examples to existence

We want to give at least two examples that are not included in the statement of Theorem 1.17 and where an optimal transport does not exist. We concentrate on examples where the cost $c(x, y)$ would usually allow for existence results (take for instance $c(x, y) = |x - y|^2$), but the reason for this lack of existence is due to the measure μ . Other examples where this is due to the cost and not to the measures are possible, see for instance Ex(58) and Ex(59).

No transport may exist The first one is very easy. Consider the case $\mu = \delta_a$, $a \in X$ and suppose that ν is not a Dirac mass. In this case there is no transport map at all. Indeed, it is easy to check that $T_{\#}\delta_a = \delta_{T(a)}$ and hence no transport map $T : X \rightarrow Y$ can exist if ν is not of the form δ_b for some $b \in Y$.

More generally, we can say that the image measure $T_{\#}\mu$ always includes an atom of mass at least $\mu(\{a\})$ for every atom a of μ . This implies in particular that measures with atoms cannot be sent through a transport map onto measures without atoms. For these reasons, the absence of atoms is a typical assumption on the source measure μ when one wants to solve (MP).

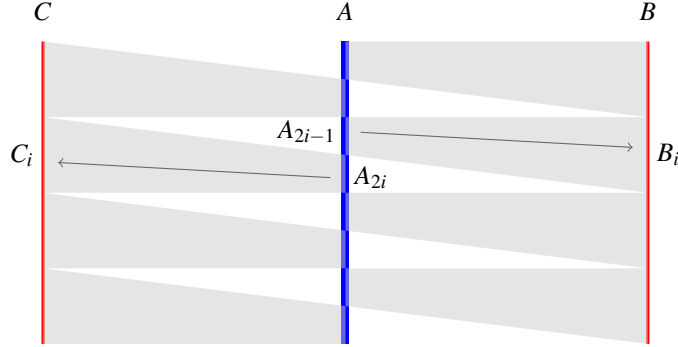
On the contrary, the problem (KP) by Kantorovich still makes sense even for atomic measures. In particular, the situation is very easy when $\mu = \delta_a$. In this case the set $\Pi(\mu, \nu)$ contains a unique element, which is $\gamma = \mu \otimes \nu = \delta_a \otimes \nu$.

Transport can exist, but no optimal one We consider the quadratic cost $c(x, y) = |x - y|^2$ (but any other cost strictly increasing in the distance would give the same result). Set

$$\mu = \mathcal{H}^1 \llcorner A \text{ and } \nu = \frac{\mathcal{H}^1 \llcorner B + \mathcal{H}^1 \llcorner C}{2}$$

where A , B and C are three vertical parallel segments in $\mathbb{R}^2$ whose vertexes lie on the two lines $y = 0$ and $y = 1$ and the abscissas are 0, 1 and -1 , respectively, and $\mathcal{H}^1$ is the 1-dimensional Haudorff measure. It is clear that no transport plan may realize a cost better than 1 since, horizontally, every point needs to be displaced of a distance 1. Moreover, one can get a sequence of maps $T_n : A \rightarrow B \cup C$ by dividing A into $2n$ equal segments $(A_i)_{i=1, \dots, 2n}$ and B and C into n segments each, $(B_i)_{i=1, \dots, n}$ and $(C_i)_{i=1, \dots, n}$ (all ordered downwards). Then define T_n as a piecewise affine map which sends A_{2i-1} onto B_i and A_{2i} onto C_i . In this way the cost of the map T_n is less than $1 + 1/n$, which implies that the infimum of the Kantorovich problem is 1, as well as the infimum on transport maps only. Yet, no map T may obtain a cost 1, as this would imply that all points are sent horizontally, but this cannot respect the push-forward constraint. On the other hand, the transport plans associated to T_n weakly converge to the transport plan $\frac{1}{2}\gamma_T^+ + \frac{1}{2}\gamma_T^-$ where $T^\pm(x) = x \pm e$ and $e = (1, 0)$. This transport plan turns out to be the only optimal transport plan and its cost is 1.

Note that in this last example we also saw that the infimum among transport maps was equal to the infimum (i.e. the minimum) among transport plans. This is a general fact, which relies on the notion of relaxation, as we can see in the following section.



1.5 Kantorovich as a relaxation of Monge

Let us set again $K(\gamma) := \int_{\Omega \times \Omega} c \, d\gamma$. Since we know that for any map T we have $M(T) = \int_{\Omega} c(x, T(x)) \, d\mu = \int_{\Omega \times \Omega} c \, d\gamma_T = K(\gamma_T)$, the Monge Problem may be re-written as

$$\min \{J(\gamma) : \gamma \in \Pi(\mu, \nu)\},$$

where

$$J(\gamma) = \begin{cases} K(\gamma) = M(T) & \text{if } \gamma = \gamma_T, \\ +\infty & \text{otherwise.} \end{cases}$$

This is simple to understand: the definition of J forces to restrict the minimization to those plan induced by a transport map. This fact is useful in order to consider (MP) and (KP) as two problems on the *same set of admissible objects*, where the only difference is the functional to be minimized, J or K .

The question is now : why did Kantorovich decide to replace J with K ? Can we easily prove that $\inf K = \inf J$? This is obviously true when, by chance, the minimizer of K is of the form $\gamma = \gamma_T$, since in this case we would have equality of the two minima. But is it possible to justify the procedure in general? The main mathematical justification comes from the following notion of *relaxation*.

Box 1.10. – Memo – Relaxation

Let $F : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a given functional on a metric space X , and suppose that it is bounded from below. We define the relaxation of F as the functional $\bar{F} : X \rightarrow \mathbb{R} \cup \{+\infty\}$ which is the maximal functional among those $G : X \rightarrow \mathbb{R} \cup \{+\infty\}$ which are lower semi-continuous and such that $G \leq F$. This functional exists since the supremum of an arbitrary family of l.s.c. functions is also l.s.c. Moreover, we also have a representation formula, which is easy to prove:

$$\bar{F}(x) = \inf \{ \liminf_n F(x_n) : x_n \rightarrow x \}.$$

A consequence of the definition is also that $\inf F = \inf \bar{F}$ (this latter infimum, that of $\bar{F}$, being often a minimum, when X is compact). This is easy to check: $F \geq \bar{F}$ implies $\inf F \geq$